

Course Syllabus

Static Timing Analysis (STA)

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Introduction:

The course program of Static Timing Analysis is assigned for undergraduate education of IC Design specialization and is taught in the 8th semester (4 year's 2nd semester).

Objective:

The goal of the course is to study STA concepts, Delay Modeling, Interconnect Parasitics and Delay Calculation. The course also focuses on configuring the STA Environment, Timing Checks and Crosstalk and Noise.

Class Hours:

The course duration is 48 hours, lectures volume is 32 hours, practice classes are 10 hours and laboratory works are 6 hours.

Homework and Exam:

Grades will be assigned on:

- Lecture (70 scores)
- Practice Classes (30 scores)
- Final exam.

Prerequisites:

The course program is compiled taking into account that the following courses had been studied in advance:

- IC Design Introduction
- Digital Integrated Circuits.

Understanding of the course is the basis for further specialized subjects destined by the educational plan of IC Design specialization.

Reference Materials:

To study the course the necessary list of references is given below.

1. V. Taraate. Digital Design from the VLSI Perspective: Concepts for VLSI Beginners. Springer; 1st ed. 2023 edition; 2022
2. K. Golshan. The Art of Timing Closure: Advanced ASIC Design Implementation. Springer; 2020
3. Z. GUANG MING . Analysis and Control of Time-Delay Singular Jump Systems. Chinese Edition. 2024
4. G. Blokdyk. Static timing analysis A Complete Guide. 5STARCook; 2018

Grading:

This course will be graded according to Professor's discretion.

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Lectures (32 hours):

Topic 1.1 (4 hours) – STA Concepts

- Static timing analysis (STA). Simulation. STA vs. simulation. Digital circuits timing goals. Problem variability. Components of circuit timing. Cell timing parameters. Cell timing constraints. Timing closure problem. Static timing analysis steps. STA concepts. Timing paths. Possible paths. False paths. Multi-cycle paths. Required time. Arrival time. Slack and critical path. Early and latest analysis. Path groups.

Topic 1.2 (4 hours) – Delay Modeling

- Path delay: Basic approach. Delay dependencies. Path delay calculation. Operating conditions. Linear delay model. Delay calculation (linear) example. Nonlinear delay model. Nonlinear delay calculation example. Two-dimensional interpolation. Nonlinear delay model (NLDM). Synopsys liberty format (.lib). Cell timing data. Cell types: sequential cells. Timing group names. Combinational timing arc syntax. Delay analysis. Timing constraints: Timing types. Hold timing example.

Topic 1.3 (4 hours) – Interconnect Parasitics

- Net. Interconnect parasitics. Overview. RC tree of interconnect. T – model. π - model. N section representation. Elmore delay. Elmore delay equation. Wire load: parasitic effects. Estimating parasitic devices. Library: wire load syntax. Library: wire load example. Library: wire load table. Interconnect trees: Best-case tree. Interconnect trees: Balanced tree. Interconnect trees: worst-case tree.

Topic 1.4 (4 hours) – Delay Calculation

- Problems in edge-triggered circuits. Required and arrival times. Path-based timing calculation. Representation of circuits. Critical path method. Given circuit with delays of components. CPM arrival time calculation steps. Required time calculation steps. Slack calculation. Slack calculation. Path slack histogram.

Topic 1.5 (6 hours) – Configuring the STA Environment

- Preparing for STA tool usage. Setting design environment. Basic clock details. Clock definition. Clock waveform. Clock uncertainty. Clock skew types. Clock jitter. Clock uncertainty modeling example. Inter-clock uncertainty. Modeling clock latency. Clock latency. Divided clock (generated). Multiplied clock (generated). Gated clock (generated). Generated by edges. Latency of generated clock. Input delay. Output delay. Input path with virtual clock. 3-cycle setup clock. 2-cycle hold check. Half-cycle path. Slow to fast clock domain. Fast to slow clock domain. Phase shifted multiple clocks. Path groups. Design externals. Driving cell. Capacitive load. Design rule checks. Combinational designs. Combinational designs. Refining. Point-to-point specification.

Topic 1.6 (2 hours) – Timing Checks

- Timing checks. Setup timing check. Launch and capture flip-flops. Data and clock signals for setup timing check. Hold timing check. Data and clock signals for hold timing check. Hold checks and setup check cycles. Removal timing check. Recovery timing check.

Topic 1.7 (4 hours) – Crosstalk and Noise

- Crosstalk. Crosstalk definition. Capacitive coupling. Noise effect types. Crosstalk delay. Crosstalk glitch. Factors affecting glitch. Types of glitches. DC noise margins. DC margin limits. AC noise limits. Basic glitch removal. Multiple aggressors. Crosstalk delay analysis. Crosstalk scenarios. Positive crosstalk. Negative crosstalk. Crosstalk timing verification. Setup analysis. Hold analysis. Crosstalk analysis for complex ICs.

Topic 1.8 (4 hours) – Statistical Static Timing Analysis

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- Variability. Process variation sources and types. Modeling variation. Process corners. Conventional corner-based analysis. Variation-aware timing analysis flow. Benefits of variation-aware analysis. Everything statistical. Statistical timing analysis. Monte-Carlo method. Introduction to statistical static timing analysis. Probability. Distribution of propagation. Statistical ADD operation. Statistical MAX operation. SSTA approaches. Spatial correlations. Spatial correlations should be considered. Considering spatial correlations. Additional data for SSTA. SSTA tool inputs and outputs.

Laboratory Works (6 hours):

Topic 2.1 (2 hours) – Generating Timing Reports

Topic 2.2 (2 hours) – Understanding Delay Calculation

Topic 2.3 (2 hours) – Domain Name System (DNS).

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